

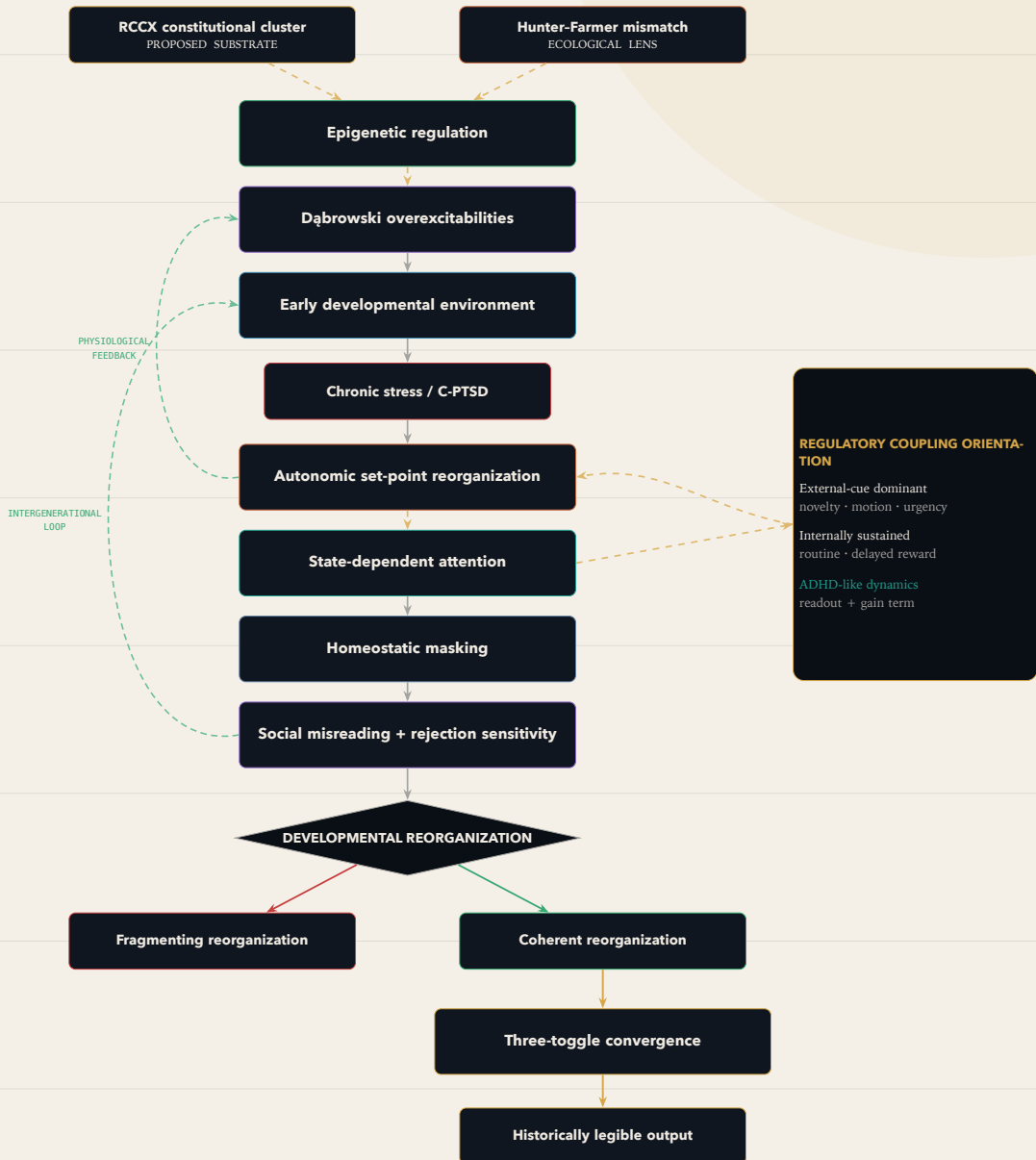
THE RCCX-HUNTER- DĄBROWSKI CASCADE

A layered model of constitutional sensitivity, regulatory coupling, developmental bifurcation, and the conversion of latent capacity into historically legible output.

The phenotype is not the diagnosis. The diagnosis is one read-out of a coupled system whose body, environment, developmental history, attention, identity, and available conversion surfaces keep acting back upon one another.

One Cascade, Multiple Certainty Levels

Nodes name layers. Badges grade claim status. Line styles distinguish directional mechanism from proposed modulation and recursive feedback. The diagram is a map of a research program, not a claim that every arrow has already been proven.



SOLID directional mechanism DASHED proposed modulation LOOPED recursive feedback

Constitution Is a Constraint, Not a Destiny

RCCX constitutional cluster

PROPOSED CLUSTER

A candidate organizing substrate, not a settled genotype-to-phenotype map.

- ▶ C4 · CYP21A2 · TNXB linkage-region proposal.
- ▶ Connective-tissue, immune, endocrine, and stress-axis convergence.
- ▶ The hypothesis concerns a constitutional probability field. It does not assign one inevitable phenotype to one genotype.

Hunter-Farmer mismatch lens

EVOLUTIONARY LENS

Ecology changes the functional value of the same regulatory tendencies.

- ▶ Broad scanning, change detection, rapid response, and active sampling may fit variable, immediate-feedback environments.
- ▶ The same tendencies may become costly inside low-variance, delayed-reward institutions.
- ▶ Hunter and Farmer are ecological renderings of a deeper regulatory architecture, not permanent human types.

Epigenetic regulation

SUPPORTED MECHANISM

Developmental environments can tune HPA-axis and glucocorticoid signaling without changing DNA sequence. Similar constitutional profiles may diverge through developmental marking. Inter-generational associations exist, but their transmission pathways remain bounded and must not be overstated.

Dąbrowski overexcitabilities

DEVELOPMENTAL THEORY

Five high-gain channels expand both possibility and effective dose:

psychomotor · sensory · emotional · intellectual · imaginal

Movement pressure, intensified sensory intake, amplified affective signal, recursive analysis, and high-throughput counterfactual generation can raise developmental potential. They can also increase the amount of environment the organism must metabolize.

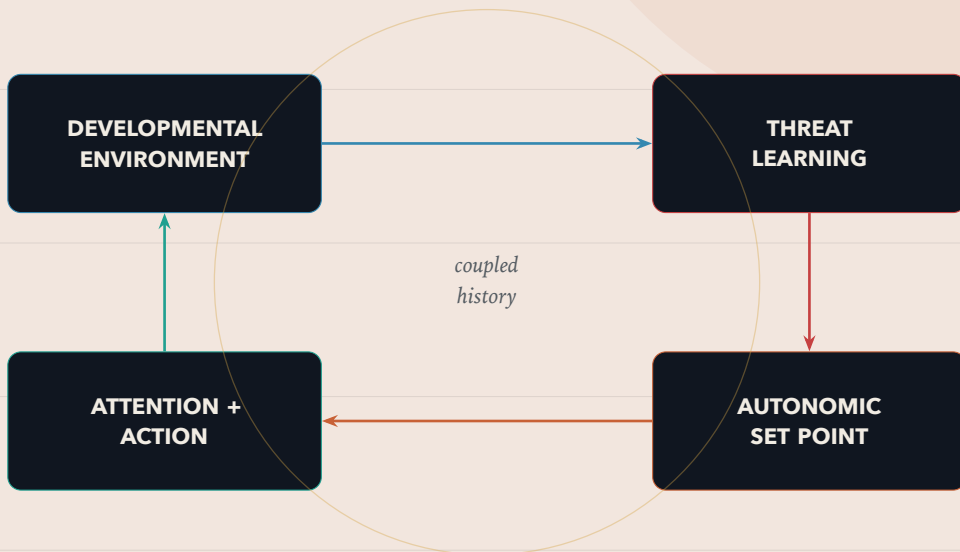
The same high-gain substrate can raise the ceiling and raise the cost of remaining organized enough to reach it.

The Body Becomes Part of the Historical Record

Early developmental environment

ESTABLISHED DOMAIN

Constitution meets reinforcement, attachment, threat, social fit, and institutional structure. Inconsistent reinforcement, emotional neglect, covert threat cues, and low-fit environments do not merely surround regulation. Environmental variance becomes part of the regulatory circuit itself.



Chronic stress / C-PTSD pathway

CLINICAL CONSTRUCT

High-gain traits can amplify the effective stress dose. Threat predictions may become durable defaults, and stress-regulatory molecular changes may persist. The claim is not that sensitivity mechanically causes trauma. It is that sensitivity, exposure, interpretation, physiology, and reinforcement can recursively change one another.

Autonomic set-point reorganization

SUPPORTED SYNTHESIS

Sympathetic bias, altered baroreflex, and changing interoceptive confidence can reorganize the conditions under which cognition operates. Volume, connective-tissue, immune, and autonomic factors may converge. POTS-like physiology can intensify cognitive and emotional state shifts. Physiology feeds upward while social context keeps writing downward.

Regulation is neither inside the organism nor outside it. It closes across the relation.

Attention Is a Readout and a Gain Term

State-dependent attention architecture

S U P P O R T E D + N O V E L F R A M E

- ▶ Stable activation can produce deep lock-in and high-resolution exploitation.
- ▶ Underactivation or threat can produce scanning, switching, and distractibility.
- ▶ Attention expresses the state of the wider stack, then feeds back through environment selection, reward, sleep timing, urgency, and accumulated consequence.

Subconscious homeostatic masking

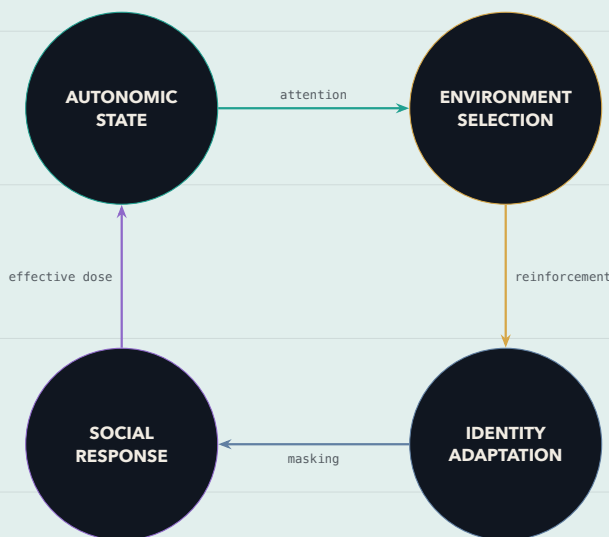
S U P P O R T E D P H E N O M E N O N

Protective routines can become automatic and identity-adjacent. Over time, adaptation, persona, and experienced self begin to blur. Camouflage varies across sex, gender, social attunement, and local threat. What looks like stable personality may partly be regulation made habitual enough to disappear from awareness.

Social misreading + rejection sensitivity

I N T E G R A T I V E O U T P U T

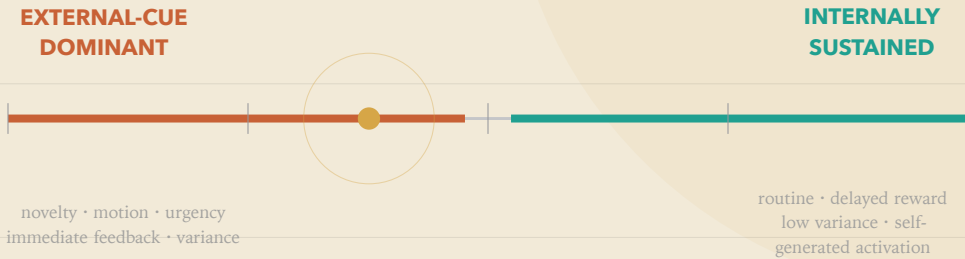
Downstream pressures are predictable without becoming character verdicts. Defensive resentment, vulnerability, and outsider identity may consolidate. The drive toward public, undeniable proof of worth can become organizing fuel. Existing frameworks feel insufficient, producing pressure to build new ones. Hyperacute room, person, and system diagnostics may be survival skill repurposed as intelligence.



The phenotype is not the diagnosis. The diagnosis is one readout of the coupled system.

Regulatory Coupling Orientation

Regulatory Coupling Orientation describes where activation and regulation are preferentially recruited from. It is a spectrum, not a Hunter-versus-Farmer binary.



Regulatory Closure Bias: the measurable index

$$R_c = \log \left(\frac{G_{\text{exo}}}{G_{\text{endo}}} \right)$$

G_{exo} is regulatory gain produced through action on and feedback from the environment. G_{endo} is gain produced through internally maintained stabilization. Positive R_c indicates exoregulatory bias; negative R_c indicates endoregulatory bias.

ADHD-like dynamics

R E A D O U T + R E C U R S I V E G A I N

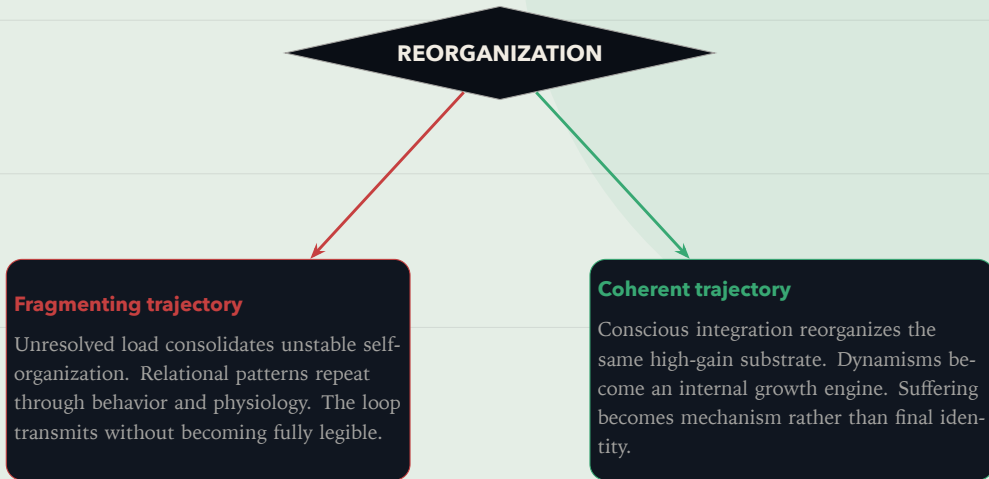
ADHD-like attention is modeled neither as sole root cause nor as decorative superpower. It can emerge from the full stack and feed back into it through environment selection, sleep timing, reward and urgency, stress exposure, autonomic state, and the consequences of prior action.

What the orientation does not mean

- ▶ It is not locus of control, blame, motivation, or conscious preference.
- ▶ It does not claim that one phenotype is universally superior.
- ▶ It does not collapse RCCX, ADHD, trauma, chronotype, or Dąbrowski into one cause.
- ▶ It predicts an interaction: phenotype × environment × available scaffolding.

What appears to be internal instability may sometimes be a regulatory system deprived of the environmental surface through which it was built to close.

The Developmental Bifurcation

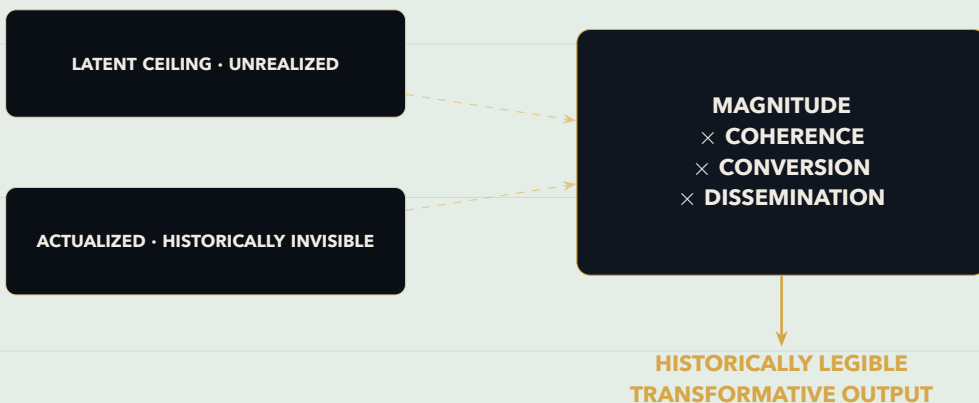


Three-toggle convergence

ORIGINAL SYNTHESIS

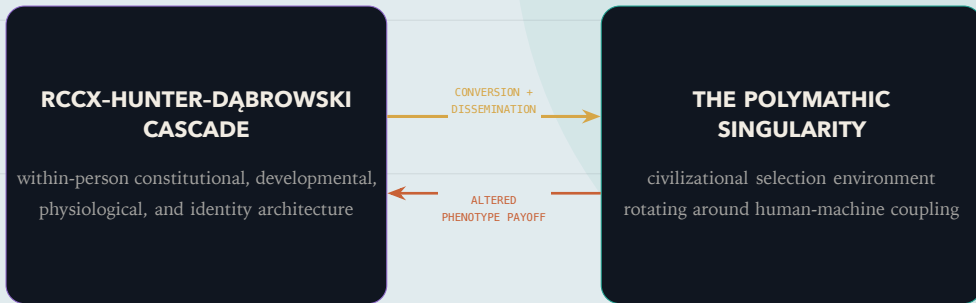
All three must align for historically legible transformative output.

- T1** **Orchestration magnitude** sets the ceiling: integrative, creative, perceptual, intellectual, emotional, and imaginative capacity.
- T2** **Reorganization direction** determines whether instability becomes fragmentation or greater coherence.
- T3a** **Capacity conversion** requires health, discipline, closure, tools, access, collaborators, and capital.
- T3b** **Dissemination** requires opportunity, uptake, networks, timing, and a channel into the historical record.



Capacity is not impact until it survives conversion and enters the historical record.

The Cascade and the Polymathic Singularity



The division of explanatory labor

The cascade models the organism and its developmental history. The Polymathic Singularity models the environment changing which configurations gain leverage, visibility, and historical consequence. The two theories couple without proving one another.

AI enters at the conversion surface, not the root substrate COUPLING CLAIM

AI can hold structure, translate among representational languages, reduce implementation cost, and expand dissemination bandwidth. It can raise conversion capacity without creating the underlying orchestration magnitude:

$$\kappa = \frac{\text{legible implemented output}}{\text{latent integrative capacity}}$$

It can also raise stimulation, branch proliferation, sleep displacement, and objective drift. The same coupling surface can amplify coherence or destabilize it.

The combined recursive loop

H O R I Z O N S Y N T H E S I S

constitutional potential → developmental organization → cognitive phenotype
→ human-AI coupling → environmental acceleration
→ altered phenotype payoff → further niche construction

The shared prediction

High-exoregulatory, integrative profiles should improve disproportionately in responsive, AI-rich, project-based environments when the system supplies enough closure to prevent amplification from becoming branch proliferation. The causal variable is not diagnosis alone. It is the fit among architecture, environment, and conversion surface.

The cascade explains how a ceiling becomes reachable. The singularity explains why the environment has begun paying for the climb.

Evidence and Source Ledger

This model is intentionally integrative. The status grammar prevents established domains, supported mechanisms, developmental theory, integrative synthesis, and original horizon claims from collapsing into one undifferentiated certainty level. These are source anchors from the v3.1 handoff, not a finished bibliography.

PROPOSED CLUSTER

Constitutional / RCCX

Meglathery, RCCX theory materials.
Castori et al., hypermobility spectrum disorders and systemic involvement.
Tinkle et al. (2017), hEDS nosology and comorbidity landscape.

EVOLUTIONARY + ATTENTION

Mismatch / state regulation

Hartmann, *Hunter in a Farmer's World*.
Sergeant (2005), cognitive-energetic model of ADHD.
Metin et al. (2012), event-rate meta-analysis.
Barack et al. (2024), ADHD traits and exploratory foraging.

SUPPORTED MECHANISM

Epigenetic stress programming

Meaney & Szyf (2005), maternal care and glucocorticoid receptor methylation.
McGowan et al. (2009), childhood abuse and hippocampal NR3C1 methylation.
Klengel et al. (2013), FKBP5 $\times$ childhood trauma.
Yehuda et al. (2015), FKBP5 methylation across generations.

DEVELOPMENTAL THEORY

Dąbrowski / reorganization

Dąbrowski (1964), *Positive Disintegration*.
Dąbrowski (1972), *Psychoneurosis Is Not an Illness*.
Mendaglio (2008, ed.), *Theory of Positive Disintegration*.

ESTABLISHED + SYNTHESIS

Attachment / stress / autonomic

Bowlby; Ainsworth; Main & Hesse, attachment foundations.
McEwen, allostatic load.
Herman (1992), complex trauma.
Thayer & Lane, neurovisceral integration.
Raj et al.; Eccles et al., POTS, hypermobility, and autonomic dysfunction.

SUPPORTED PHENOMENON

Masking / social adaptation

Jung, persona and shadow.
Cage et al. (2018), autistic camouflaging.
Hull et al. (2019), CAT-Q validation.
Lai et al., sex/gender differences in autism presentation.

HORIZON DIRECTION

Chronotype / coupling

Samson et al. (2017), chronotype diversity and nighttime coverage among Hadza.
Coogan & McGowan (2017), ADHD circadian function and eveningness.
Yetish et al. (2015), sleep in three preindustrial societies.

NEXT FORMALIZATION TARGET Operationalize Regulatory Coupling Orientation independently of diagnosis; specify G_{exo} and G_{endo} across task environments; distinguish causal, modulatory, and recursive links; preregister phenotype $\times$ environment predictions; test when AI raises κ and when it raises fragmentation instead.

T H E C A S C A D E D O E S N O T E N D A T
T H E D I A G N O S I S

THE SAME SUBSTRATE CAN BECOME A CAGE, A COMPASS, OR A CEILING.

Constitution changes the probability landscape. Development writes into regulation. Regulation reorganizes attention and action. Social response becomes additional environment. The system then reaches a real bifurcation: not between gifted and disordered, Hunter and Farmer, or sensitive and strong, but between trajectories that remain trapped inside their own recursive load and trajectories that learn to convert that load into coherence. Even coherence is not enough. Capacity must survive conversion. Output must survive dissemination. The historical record does not contain every mind that reached the ceiling. It contains the minds whose ceilings acquired a surface through which they could become legible.



CONSTITUTION → REGULATION → REORGANIZATION
→ HISTORICAL LEGIBILITY